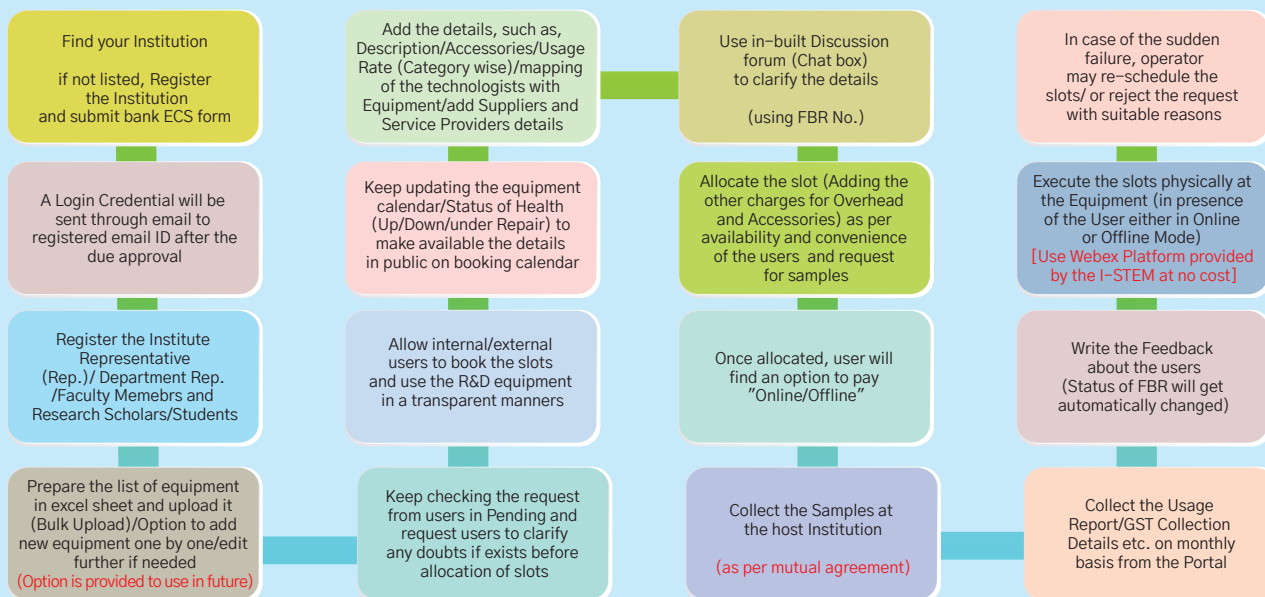




Guidelines for Custodian Organization (Academia and Industry)



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Migration from NOR to NAND

To shift from NOR flash to NAND flash memory, some partitions of 1Gb serial NAND memory can be

done. Traditionally, the NOR flash could be 128kB or even 1MB. So, with 128MB, nearly 256 blocks can be put with an additional 32MB. It now becomes a boot-code firmware,

RTOS, or some application software.

Maybe, one-quarter of that is good enough for some embedded system in your device. Partition it so that the rest can be used for user data or the system log data. Some areas also have to be reserved for driver use.

The NAND block may contain some bad blocks, meaning, some blocks could be damaged. So, the reserve area can replace the bad blocks.

When you migrate from NOR to NAND flash, a software driver is required, which could include a serial peripheral interface (SPI). As mentioned earlier, NAND is suitable for applications with intensive write/erase requirements or storage, such as smart-

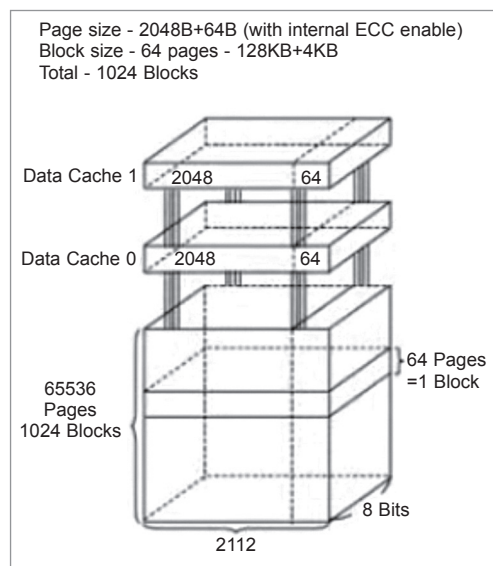


Fig. 1: 1Gb NAND flash memory structure (Credit: KIOXIA)

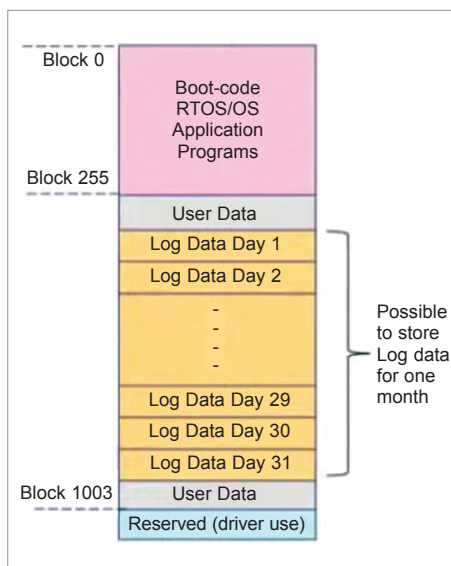


Fig. 2: Example of 1Gb NAND flash partition